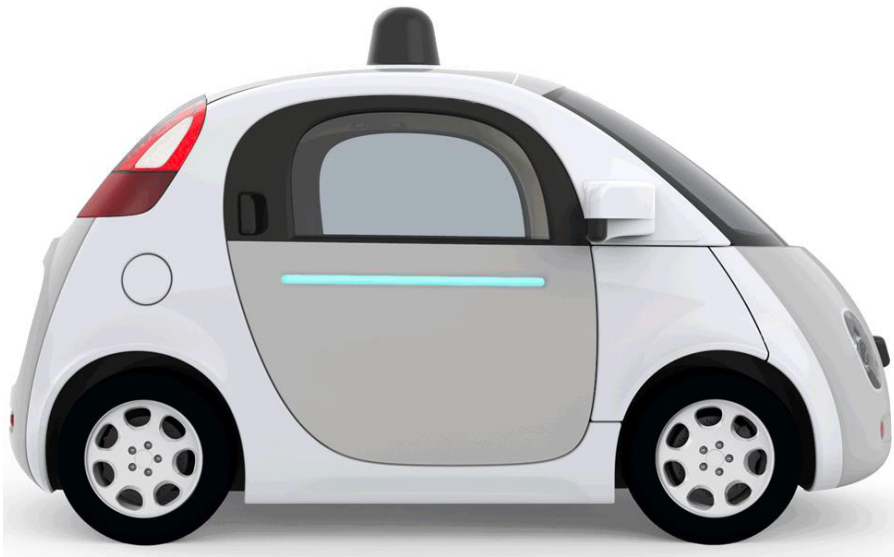


Embedded Electronic Engineering Lab A

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► Introduction

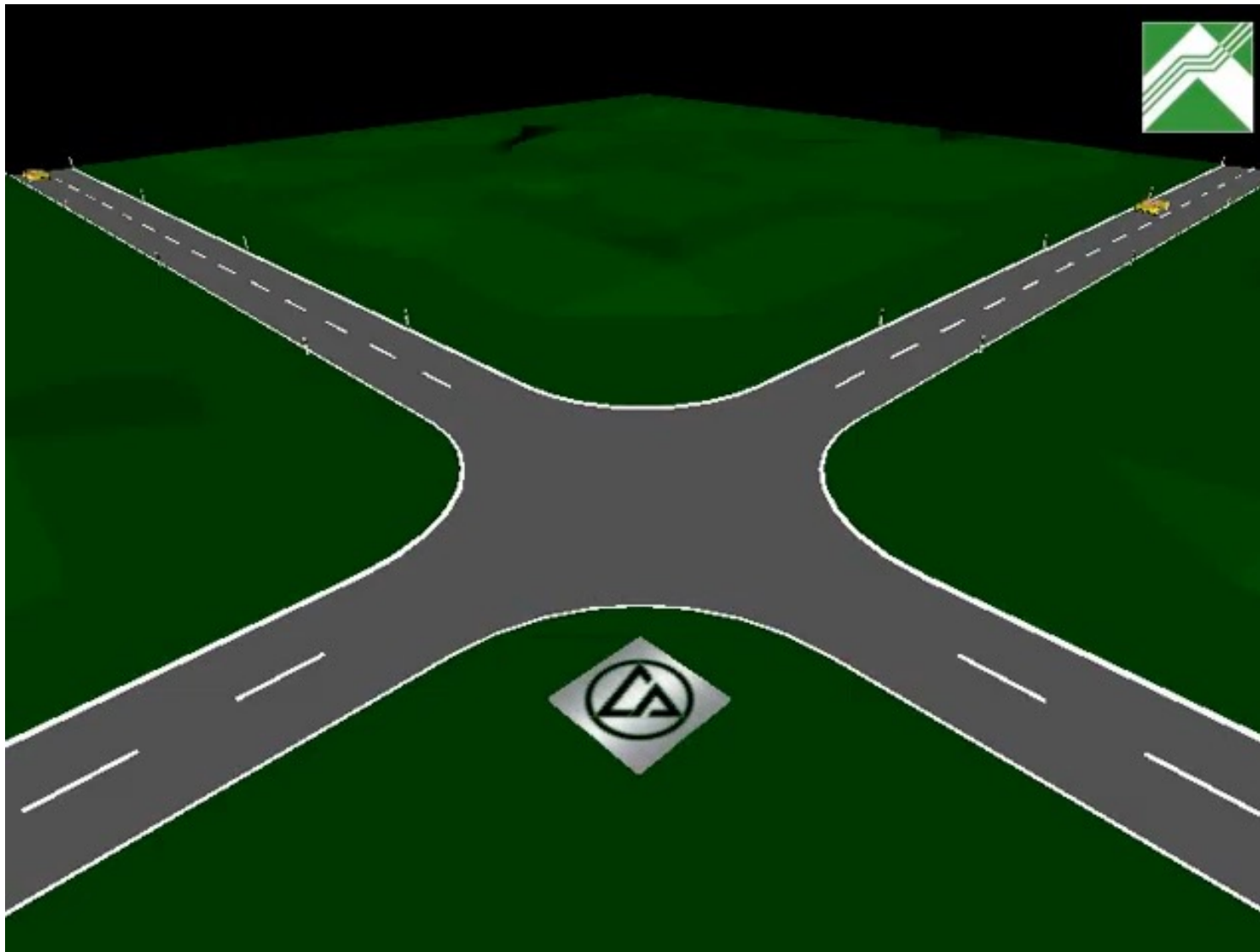


© Googles self-driving car (Waymo)



Cross-Traffic Management © IMK Engineering

► Introduction



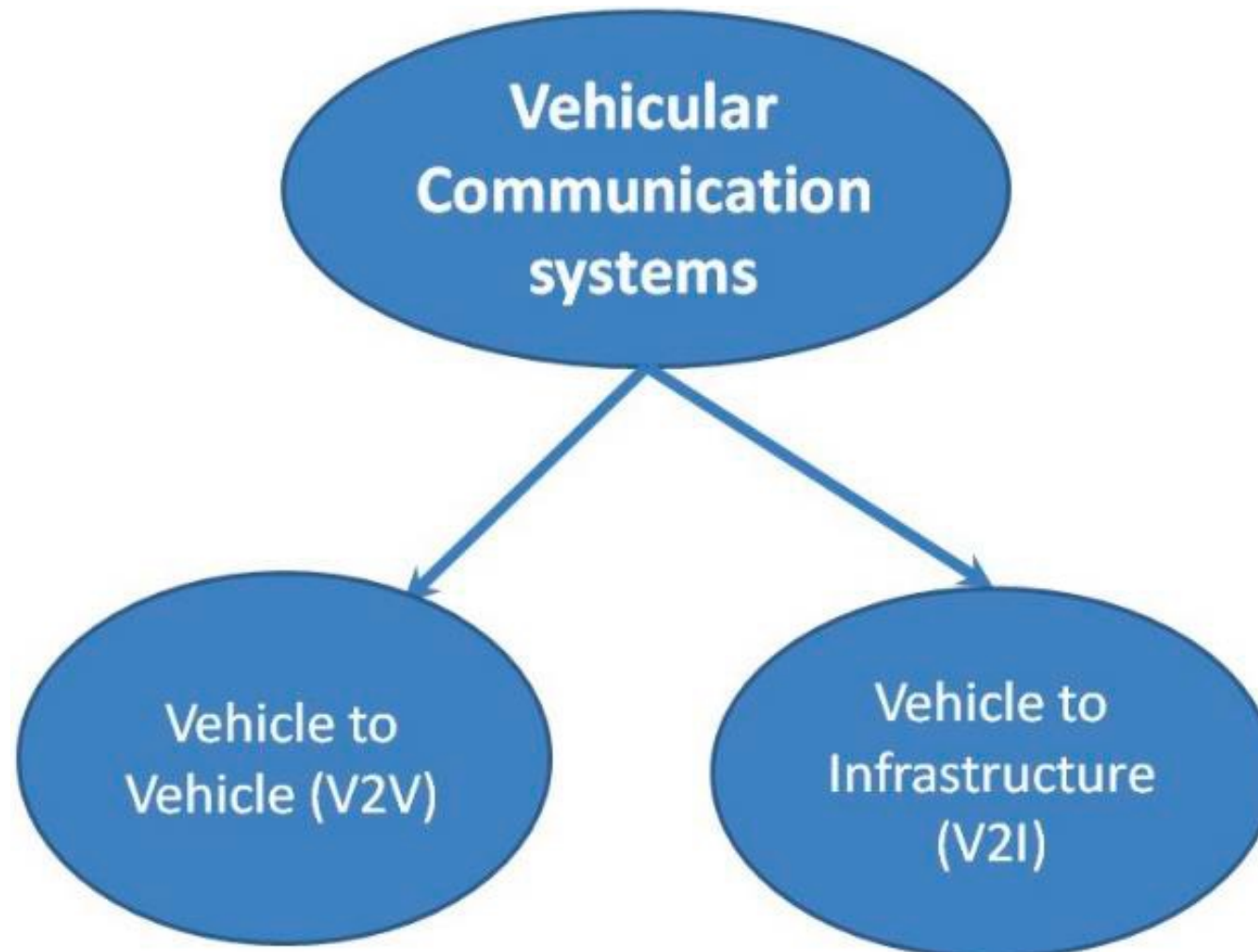
Cross-Traffic Management © IMK Engineering

To achieve the objectives of an efficient cross-traffic management, the work is divided to parts to address the following research questions:

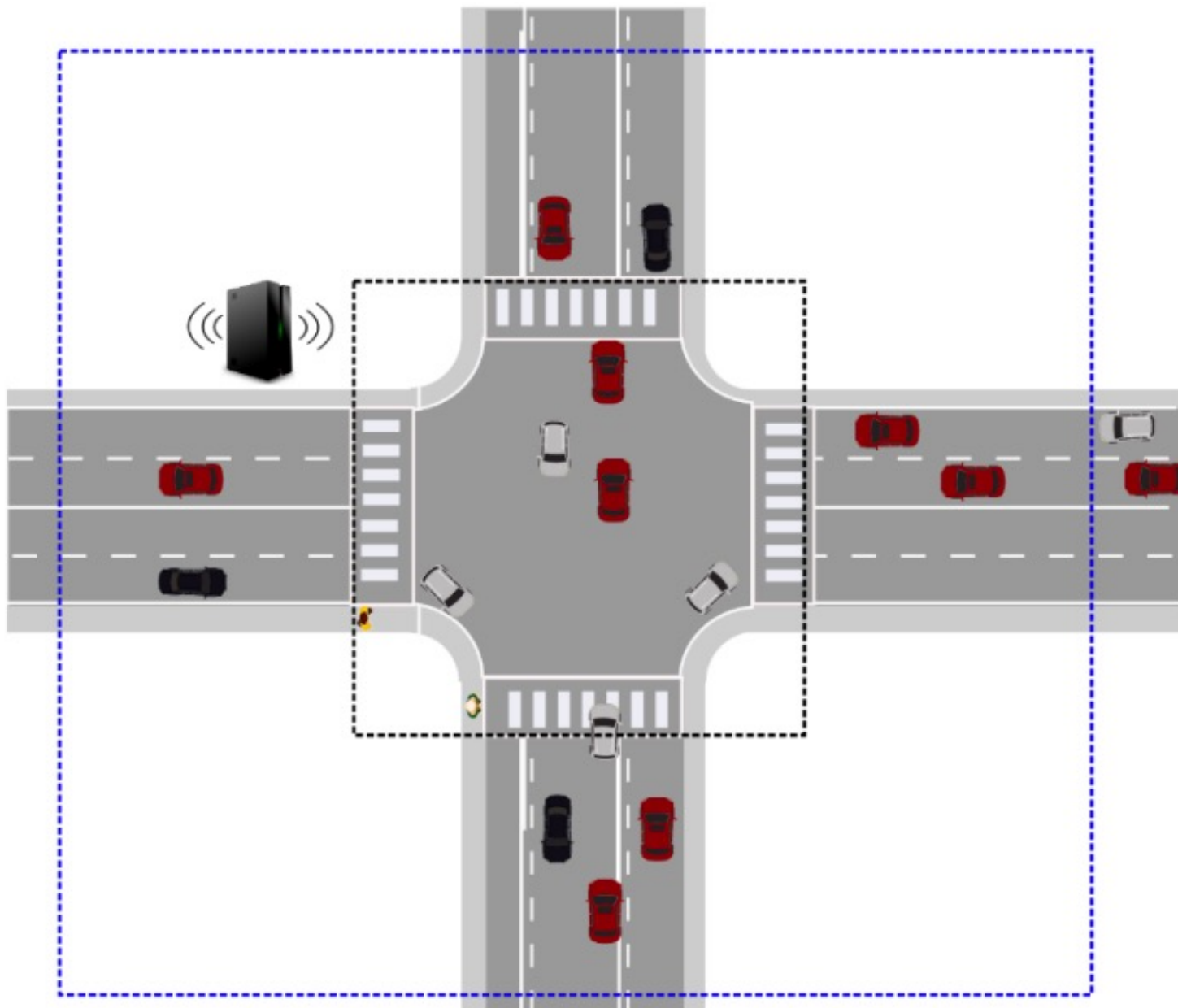
- How to model a traffic in intersection for autonomous cars using Queuing system?
- What is the best scenario to reduce the traffic congestion in an intersection?
- How to simulate traffic intersection using C-code (core logic) and freeRTOS, and what simulation parameters need to be considered?

- Design Methodology
 - Select an appropriate methodology you learned in the past classes and follow this approach
- Queuing theories
 - Everybody faced queues many times during our normal day, just like when somebody going to bank to draw or deposit money, he have to wait in a queue in order to be served by the service desk, people who arrive earliest have to be served first that kind of queues and serving system called (FIFS first in first served), if there were only one service desk the queuing and serving process will be classified as M/M/1 in general we can identify having one arrival and one server at a time can be modelled as an M/M/1 queuing system, if the servers were two then the modelling will be M/M/2, if they are three then M/M/3 and so on. In our case we will study the M/M/1 and M/M/2.

- Communication of autonomous vehicles



► Cross-traffic Example



- Team building
 - Each group has 19 members
 - 4 teams with 4 members, 1 team with 3 members
- Work with git
 - Create a github repository for your team
 - Invite Profs (shenkler, rettberg) and research assistants (Krypto-Klaus)
 - Deadline May 17 during our session – ideally earlier handled
- Upload continuously your results to git
 - These includes the responsibilities
 - (Pre-) final version are uploaded within of the specified deadlines
- Divide the overall task into separate parts for each team-member in the following way, like:
- Or via a git project

			Name1		Name2		Name...
#	Task	Short summary	Todo (incl. Deadline)	Done (incl. Finishing date)	Todo	Done	...
1	Task1						
2	Task2						
...	Task...						

- Requirements
 - Use case diagram
- Modelling
 - UPPAAL Timed Automata
 - If relevant sequence diagrams, activity diagrams, ...
- Implementation C-code with freeRTOS (<https://www.freertos.org/>)
 - Well-defined mapping from models
- Setup a Github repository
- Everyone will write a report of his/her project that will be a chapter of the final report

- Use github
- Milestones paper & implementation
 - We start from May 18 with counting the weeks. Deadline is always the Sunday before our next session. Each session is used as a weekly meeting for the teams. Together with the profs the current status is discussed and new tasks are defined.
 - Week 1: requirements and use-cases
 - Week 2: show models and diagrams
 - Week 3: first abstract implementation
 - Week 4: proposal for HW/SW codesign
 - Week 5: codesign implementation
 - SW with freeRTOS
 - HW with ModelSim <https://www.intel.de/content/www/de/de/software/programmable/quartus-prime/model-sim.html>
 - Last (penultimate) week of semester: final version & presentation (20 min talk + 10 min discussion)
- Paper
 - Deadline for submission: end of first week of exam phase, eob

- Use latex and bibtex
- See ieee template at: https://www.ieee.org/content/dam/ieee-org/ieee/web/org/pubs/conference-latex-template_10-17-19.zip
- 2 to 4 pages per team member
- Apply the methods and techniques from scientific writing
- Use citavi or Zotero for managing your references
- Write for each related reference a short note of how it is related to your seminar topic – the note is part of your bibtex management tool and/or an extra bibtex entry.

- The evaluation of each special emphasis module is given by
 - Two seminars – each 1/3 of overall evaluation -> at all 2/3
 - Lab 1/3 of overall evaluation
- In cases of the lab + seminar
 - The paper, talk and implementation will both be considered all